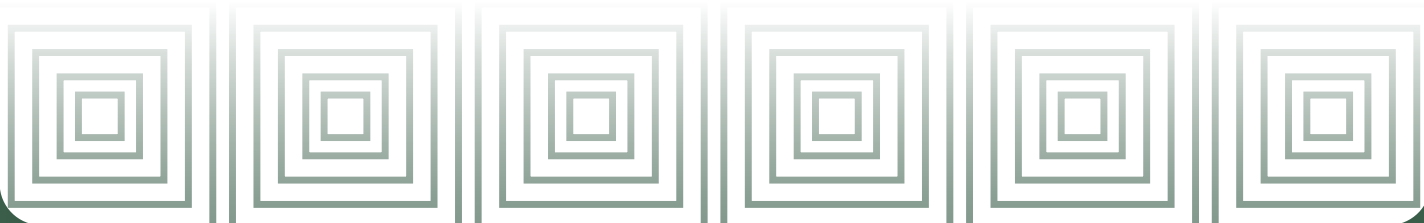


BASIC MATHEMATICS SYLLABUS

FOR ADVANCED LEVEL (S4 - S6)

LEARNING PATHWAYS: ARTS AND HUMANITIES; LANGUAGES

Kigali, 2026



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FOREWORD

The Rwanda Basic Education Board (REB) is honoured to present the Basic Mathematics syllabus (S4-S6) of Arts and Humanities and Languages Learning pathways. It serves as an official guide for competence-based teaching and learning, ensuring consistency in delivering quality education across the country. Developed collaboratively by subject specialists, educators, and curriculum developers, it supports the implementation of the Competence-Based Curriculum (CBC) and reflects Rwanda's commitment to nurturing learners' knowledge, skills, values, and attitudes.

Quality education is the foundation of Rwanda's sustainable development and prosperity. The underlying principle behind the introduction of this subject is to ensure that the curriculum responds to the needs of the Learners, the society, and the labour market. Mathematics is one of subjects of Competence Based Curriculum that contributes to shape the Learners with required knowledge, skills, attitudes, and values to produce well-trained Learners at Secondary school level. Aligned with the national education philosophy, this syllabus promotes critical thinking, creativity, inclusiveness, and holistic learner development. It aims to empower young people with competencies necessary to achieve full potential, equipping them with relevant knowledge, skills, the right attitudes and values to integrate effectively into society and pursue opportunities in a competitive world.

Anchored in Rwanda Vision 2050 and the National Strategy for Transformation (NST2), this syllabus supports the nation's goal of building a knowledge-based economy driven by innovation, resilience, and global competitiveness. It prioritizes learner-centred pedagogy, active learning, and the practical application of knowledge to prepare student not only for academic success but also to become responsible, ethical, and productive citizens capable of contributing to Rwanda's continued transformation.

This syllabus reflects our determination to equip learners with 21st-century skills, foster innovation, and instil values of integrity, social responsibility, and lifelong learning. Its success depends largely on the dedication and professionalism of teachers and school leaders. We therefore call upon all educators to embrace the approaches, methodologies, and assessment strategies outlined herein with commitment and passion, transforming classrooms into dynamic spaces of curiosity, creativity, and inspiration.

On behalf of REB, I extend my heartfelt appreciation to all who contributed to the development of this syllabus. Your expertise and dedication have been instrumental in shaping a curriculum that responds to the needs of our learners and supports.

Together, we can ensure that education continues to be a powerful driver of our nation's development and prosperity.



Dr. MBARUSHIMANA Nelson

Director General

ACKNOWLEDGEMENT

The development of this syllabus has been a collective effort that reflects the dedication, expertise, and commitment of many stakeholders. On behalf of the Curriculum, Teaching and Learning Resources Department of the Rwanda Basic Education Board (REB), I wish to express my deepest appreciation to all who contributed to the successful preparation of this Basic Mathematics syllabus for Arts and Humanities and Languages Learning Pathways.

Special thanks go to the subject experts, curriculum developers, teachers, and education partners whose insights, feedback, and professional contributions have been invaluable in shaping this syllabus. Your collaboration has ensured that the syllabus is not only aligned with Rwanda's Competence-Based Curriculum but also responsive to the aspirations of Vision 2050 and the National Strategy for Transformation (NST2).

We also acknowledge the invaluable support of development partners and education stakeholders who have consistently provided technical assistance in curriculum adaptation. Your partnership has greatly contributed to strengthening the quality of Rwanda's education system.

Finally, we call upon teachers, school leaders, and all education practitioners to use this syllabus with commitment and professionalism, ensuring that classrooms become vibrant learning environments that nurture knowledge, skills, values, and positive attitudes among learners. Through your dedication, this syllabus will fulfil its purpose of empowering learners to realize their full potential and contribute meaningfully to Rwanda's sustainable development.



MURUNGI Joan

Head of Curriculum Teaching and Learning Resources Department/REB

TABLE OF CONTENTS

FOREWORD	iii
ACKNOWLEDGEMENT	v
1. INTRODUCTION	1
1.1 Background to syllabus adaptation	1
1.2 Rationale of teaching and learning mathematics	1
1.3 Pedagogical approach	5
1.4 Assessment approach	8
1.5 Resources.....	12
2. SYLLABUS UNITS	18
2.1 Presentation of the structure of the syllabus units	18
2.2 Basic Mathematics units for Senior Four	20
2.3 Mathematics units for Senior Five	29
2.4 Mathematics syllabus units for Senior Six.....	41
BIBLIOGRAPHY	51
ANNEX: TIME ALLOCATION PER SUBJECT IN CONCERNED PATHWAYS	53



Experimental version

1. INTRODUCTION

1.1 Background to syllabus adaptation

The adaptation of the Basic Mathematics syllabus for Upper Secondary school reflects Rwanda's ongoing commitment to strengthening the Competence-Based Curriculum (CBC), first rolled out in 2015. Continuous feedback from teachers, school leaders, and education partners highlighted the need for improvements in content delivery, time allocation, and responsiveness to learner needs.

This syllabus incorporates the Ministerial directive to harmonize school hours to five per day and 33 weeks of learning annually and ensures greater coherence, inclusiveness, and alignment with national priorities such as *Vision 2050* and the *National Strategy for Transformation (NST2)*, which emphasize a knowledge-based economy, responsible citizenship, and sustainable development.

Basic Mathematics syllabus guides the interaction between the teacher and the learner through the learning processes and highlights the essential practical skills and competences a student should acquire during and at the end of each unit of learning.

1.2 Rationale of teaching and learning mathematics

1.2.1 Mathematics and society

The Basic Mathematics syllabus has put emphasis on integrated production skills and on an integrated approach in all disciplines. Mathematics is an excellent vehicle for the development and improvement of a person's intellectual competence in logical reasoning, spatial visualization, analysis and abstract thought. Learning mathematics develops numeracy, logical reasoning skills, critical thinking skills and problem-solving skills. This will result in mathematics being used in many activities of daily life thereby serving as an important tool to the society. In this way the subject will be demystified and user friendly.

Therefore, mathematics plays an important role in society through abstraction and logic, counting, calculation,

measurement, systematic study of shapes and motion. It is also used in natural sciences, engineering, medicine, finance and social sciences. Applied Mathematics like statistics and probability play an important role in game theory, in the national census process, in scientific research, etc. In addition, some cross-cutting issues such as financial awareness are incorporated into some of the mathematics units to improve the social and economic welfare of Rwandan society.

1.2.2 Mathematics and learners

Learners need enough basic mathematical competences to be effective members of Rwandan society, including the ability to count, estimate, measure, calculate, handle and manage money, interpret statistics, assess probabilities, and read commonly used mathematical representations and graphs. Reading or listening to the news requires many of these competences and citizenship requires being able to interpret critically the information one receives. For example, understanding an age-length or age-weight graph helps parents and health practitioners monitor the health of a child.

Mathematics also equips learners with knowledge, skills and attitudes necessary to enable them to succeed in an era of rapid technological growth and socio-economic development. Mastery of basic mathematical ideas and operations should make learners confident in problem-solving in life situations. A high-quality mathematics education therefore provides a foundation for understanding the world, the ability to reason mathematically, an appreciation of the beauty and power of mathematics, and a sense of enjoyment and curiosity about Mathematics subject.

In this syllabus the teacher has the task of trying to make mathematics a reality in life. Methods and approaches to learning experiences should be mostly practical and based on the experience of the learners. Hence, teaching methods to be emphasized are those that allow learners to explore, try different procedures and solve problems practically. Learning mathematics needs to include practical problem-solving activities with opportunities for students to plan their own investigations and develop their mathematical competency and confidence. Moreover, as new technologies have had a dramatic impact on all aspects of life, wherever possible in mathematics, learners should gain experience of a range of ICT applications.

1.2.3 Competences

Competence is defined as the ability to perform a particular task successfully, resulting from having gained an appropriate combination of knowledge, skills and attitudes.

The mathematics syllabus provides the opportunity for learners to develop different competencies, including the generic competencies.

Basic competencies are addressed in the stated broad subject competences and in objectives highlighted year on year basis and in each units of learning. The generic competencies, basic competences that must be emphasized and reflected in the learning process are briefly described below and teachers will ensure that learners are exposed to tasks that help the learners acquire the skills.

Generic competences and values

- **Critical and problem-solving skills:** Learners use different techniques to solve mathematical problems related to real life situations. They are engaged in mathematical thinking, they construct, symbolize, apply, and generalize ideas.

The acquisition of such skills will help learners to think imaginatively and broadly to evaluate and find solutions to problems encountered in all situations.

- **Creativity and innovation:** The acquisition of such skills will help learners to take initiatives and use imagination beyond the knowledge provided to generate new ideas and construct new concepts. Learners will improve these skills through mathematics contests, and mathematics competitions, etc.
- **Research:** This will help learners find answers to questions based on existing information and concepts as well as explain phenomena based on findings from information gathered.
- **Communication in official languages:** Learners communicate effectively their findings through explanations, construction of arguments and drawing relevant conclusions.

Mathematics teachers, irrespective of not being teachers of language, will ensure the proper use of the language of instruction by learners. This will help learners communicate clearly and confidently and convey ideas effectively through speaking and writing and use the correct language structure and relevant vocabulary.

- **Cooperation, interpersonal management and life skills:** Learners are engaged in cooperative learning groups to promote higher achievement rather than competitive and individual work.

This will help learners to co-operate with others as a team in whatever task are assigned and to practice positive ethical moral values and respect for the rights, feelings and views of others. Learners will perform practical activities related to environmental conservation and protection. They will also advocate for personal, family and community health, hygiene and nutrition and respond creatively to the variety of challenges encountered in life.

- **Lifelong learning:** The acquisition of such skills will help learners update their knowledge and skills with minimum external support and to cope with the evolution of advances in knowledge for personal fulfilment in areas that need improvement and development.

Broad Basic Mathematics Competences

Upon completion of the S4-S6 cycle for Non-Scientific Pathways, the student will be able to:

- Solve real-world problems by applying basic mathematical knowledge and skills.
- Articulate the role and value of mathematics in social, economic, and daily life contexts.
- Apply statistical skills to process, interpret, and evaluate data from reports, media, and other sources for informed decision-making.
- Construct logical arguments and use quantitative reasoning to justify decisions and solve practical problems.
- Utilize digital tools and AI to explore mathematical concepts and find solutions.

- Show a foundation in Mathematics for further studies in social sciences and related fields at the tertiary level.

Mathematics and developing competences

The national policy documents based on national aspirations identify some 'Basic Competencies' alongside the 'Generic Competencies' that will develop higher order thinking skills and help students learn subject content and promote the application of acquired knowledge and skills.

Through observations, constructions, hands-on manipulations, generalizations, and presentations of information during the learning process, the learner will not only develop deductive and inductive skills but also acquire co-operation, communication, critical thinking and problem-solving skills. This will be realized when learners make presentations leading to inferences and conclusions at the end of the learning unit. This will be achieved through group work activities and co-operative learning which in turn will promote interpersonal relations and teamwork.

The acquired knowledge in learning mathematics should develop a responsible citizen who adapts to scientific reasoning and attitudes and develops confidence in reasoning independently.

1.3 Pedagogical approach

The change to a competence-based curriculum is about transforming learning to ensure that learning is deep, enjoyable and habit-forming.

IT in general and particularly ICT should be used as a pedagogical tool to facilitate teaching and learning of mathematics.

Various teaching strategies and approaches such as direct instruction, discovery learning, investigation, guided discovery or other methods must be incorporated. Among the approaches that can be given consideration include the following:

- Learner-cantered learning.
- Different learning abilities and styles of learners (individualization);
- Co-operative learning.
- Contextual learning.
- Mastery learning.
- Constructivism.
- Use of relevant, suitable and effective teaching materials.
- Formative evaluation to determine the effectiveness of teaching and learning processes.

The choice of a suitable approach will stimulate the teaching and learning environment inside or outside the classroom.

1.3.1 The role of the learner

In the competence-based curriculum, the learner is the principal actor of his/her education. He/she is not an empty bottle to fill. Taking into account the initial capacities and abilities of the learner, the activities of the learner are indicated against each learning unit and reflect appropriate engagement of the learner in the learning process. The teaching-learning process will be tailored towards creating a learner friendly environment based on capabilities, needs, experience and interests.

Therefore, the following are some of the roles or expectations from learners:

- Learners construct the knowledge either individually or in groups in an active way. From the learning theory, learners move in their understanding from concrete through to pictorial to abstract. Therefore, the opportunities should be given to learners to manipulate concrete objects and to use models.
- Learners will be encouraged to do research and present their findings through group work activities.

- A learner is co-operative: learners work in heterogeneous groups to increase tolerance and understanding.
- Learners are responsible for their own participation and for making sure others participate.
- Help is sought from within the group and the teacher is asked for help only when the whole group agrees to ask a question.
- Consensus on the answer is required from the whole group.
- The group evaluates its own strategies and ideas rather than relying on the teacher for this evaluation.
- The learners who learn at a faster pace do not do the task alone and then the others merely sign off on it.
- Participants ensure the effective contribution of each member, through clear explanation and articulation of constructive arguments, to improve their English literacy, develop a sense of responsibility and to increase their self-confidence, and public speaking ability, etc.

1.3.2 The role of the teacher

Some of the specific duties of the teacher when implementing competence-based activities are as follows:

- He/she is a facilitator: his/her role is to provide opportunities for learners to meet problems that create interest and challenge them and that, with appropriate effort, they can solve.
- He/she is an organizer: his/her role is to organize the learners, in the classroom or outside, and engage them through participatory and interactive methods through the learning processes as individuals, in pairs or in groups. To ensure the learning is personalized and active and participative, the teacher must identify the needs of the learners, the nature of the learning to be done, and the means to shape learning experiences accordingly.
- He/she is an advisor: he/she provides counselling and guidance for learners in need. He/she comforts and encourages learners by valuing their contributions in the class activities.

- He/she is a conflict-solver: when members of a group have problems such as the attribution of tasks he/she should provide useful and constructive ideas. The teacher should settle disputes among the group.
- He/she is ethical: he/she teaches by example, by being impartial, by being a role-model, and by caring for individual needs, especially for slow learners and learners with special educational needs.

1.3.3 Special needs education and inclusive approach

All Rwandans have the right to access education regardless of their different needs. The underpinnings of this provision would naturally hold that all citizens benefit from the same menu of educational programs. The possibility of this assumption is the focus of special needs education. The critical issue is that we have persons/learners who are totally different in their ways of living and learning as opposed to the majority. The difference can either be emotional, physical, sensory and/or intellectual learning challenged, traditionally known as mental retardation.

These learners equally have the right to benefit from the free and compulsory basic education in nearby ordinary/mainstream schools. Therefore, the schools' role is to enrol them and also set strategies to provide relevant education for them. The teacher therefore is requested to consider each learner's needs during the teaching and learning process. Assessment strategies and conditions should also be standardized to the needs of these learners. Detailed guidance for each category of learners with special education needs is provided for in the guidance for teachers.

1.4 Assessment approach

Assessment evaluates the teaching and learning process through collecting and interpreting evidence of an individual learner's learning progress and makes a judgment about the learner's achievements measured against defined standards. Assessment is an integral part of the teaching learning process. In the new competence-based curriculum, assessment must also be competence-based, whereby a learner is given a complex situation related to his/her everyday life and asked to try to overcome the situation by applying what he/she has learned.

Assessment will be organized at the following levels: School-based assessment, District examinations, Learning Achievement in Rwandan Schools (LARS), and National examinations.

1.4.1 Levels of assessment in Advanced Level across terms and national examinations

To provide a comprehensive picture of learner achievement and development, assessment is conducted at multiple levels:

- i) **Diagnostic:** Conducted before instruction to identify learners' prior knowledge and skills; administered at the beginning of a unit or lesson to guide teaching.
- ii) **Formative / Continuous:** Carried out during the learning process to track progress and provide feedback; implemented continuously through quizzes, projects, class activities, and teacher observations.
- iii) **Summative assessment:**
 - **End Term I:** Administered after instruction to evaluate learners' achievement of Term I content; conducted at the end of the first term as a school-based assessment.
 - **End Term II:** District-coordinated assessment designed to measure achievement of Term II content; developed by schools and quality-checked at district level.
 - **End Term III:** National assessment to evaluate mastery of the entire year's content; prepared, set, and moderated by NESA at the end of the third term.
 - **National Examinations (NESA):** High-stakes assessments for certification, selection, and accountability; administered nationally at the end of key education cycles (P6, S3, S6) under NESA's authority. This examination integrates objective questions, structured responses, and applied tasks aligned with CBC principles to holistically measure learner achievement.
 - In Advanced Level, all students will take 11 subjects in total – 7 core subjects and 4 compulsory subjects. The compulsory subjects include ICT, Entrepreneurship, General and Religious Studies, and Physical Education and Sport, which are mandatory for all learners across all pathways.
 - All subjects will be assessed at the national level, except for Physical Education and Sport, General and Religious Studies, ICT, and Entrepreneurship, which will be assessed at the school level only.

Average scores from school, district, and comprehensive summative assessments will be weighted and included in the final national examination grading, initially contributing 10%, with the percentage gradually increasing as teachers gain experience and confidence in assessment practices.

1.4.2 Record keeping

Accurate record keeping is essential for monitoring learner progress and informing instructional decisions. Teachers use tools such as portfolios, checklists, and report books to collect evidence of learning. Portfolios include samples of learners' work, test results, projects, and self-reflections, serving as both a record of achievement and a diagnostic tool for guiding remedial action. These records provide data for feedback to learners, teachers, and parents, and contribute to final evaluations.

1.4.3 Item writing in summative assessment

Summative assessment items must be valid, fair, reliable, and aligned with CBC principles. They should test a range of competencies, including knowledge recall, application, analysis, and values integration, while reflecting real-life contexts. Item writers follow a structured process that involves:

- Identify topic areas to be tested from the subject syllabus.
- Outline subject matter content to be considered as the basis for the test.
- Identify learning outcomes to be measured by the test.
- Preparing a test blueprint or table of specifications;
- Using verbs that require higher-order thinking beyond memorization;
- Applying transparent and consistent marking schemes.
- Greater weight is given to higher-order cognitive skills as stated in the syllabus.

1.4.4 Structure and format of national examinations

There will be one paper in Basic Mathematics as a Minor Core Subject at the end of Secondary. The paper will be composed of two sections, where the first section will be composed with short answer items or items with short calculations which include the questions testing for knowledge and understanding, investigation of patterns, quick calculations and applications of Mathematics in real life situations. The second section will be composed with long answer items or answers with demonstrations, constructions, high level reasoning, analysis, interpretation and drawing conclusions, investigation of patterns and generalisation.

The items for the second section will emphasise the mastering of mathematical facts, the understanding of mathematical concepts and its applications in real life situations. In this section, the assessment will find out not only what skills and facts have been mastered, but also how well learners understand the process of solving a mathematical problem and whether they can link the application of what they have learned to the context or to the real-life situation. The time required for the paper is three hours (3hrs).

The following topic areas must be assessed: Algebra; calculus; geometry; statistics and probability. In the second section learners should have the right to choose to answer 3 items out of 5.

1.4.5 Reporting to parents

Regular communication with parents is an essential component of the CBC. This is achieved through progress reports, parent-teacher meetings, and school community activities. Since the CBC emphasizes a wide range of learning outcomes: knowledge, skills, attitudes, and competences a single mark is insufficient to capture learner progress. Effective reporting should highlight areas where learners are excelling as well as those requiring improvement. Using descriptive feedback or simple rating scales (e.g., meeting expectations very well, meeting expectations, not meeting expectations) across knowledge, subject-specific skills, and competences provides a clearer and more comprehensive picture than numerical grades alone. This approach fosters meaningful parental involvement and supports learners' continued growth.

1.5 Resources

1.5.1 Materials needed for implementation

The following list shows the main materials and equipment needed in the learning and teaching process:

- Materials to encourage group work activities and presentations: Computers (desktops and laptops) and projectors; manila papers and markers
- Materials for drawing and measuring geometrical figures/shapes and graphs: Geometric instruments, ICT tools such as GeoGebra, Microsoft student ENCARTA
- Materials for enhancing research skills: Textbooks and internet (the list of the textbooks to consult is given in the reference at the end of the syllabus and those books can be found in printed or digital copies)
- Materials to encourage the development of mathematical models: scientific calculators, Math type, MATLAB, etc.
- Artificial Intelligence (AI) should be used as a tool to facilitate the lesson planning, activity setting, scheme of work preparation and the verification of answers for some activities that require many steps.

Note that the technology used in the teaching and learning of Mathematics should be regarded as tools to enhance the teaching and learning process and not to replace teachers.

1.5.2 Use of AI in Mathematics teaching and learning

AI became a powerful ally in mathematics education when used thoughtfully. The key is viewing AI as a powerful tool in our educational toolkit, not as a threat to genuine mathematical understanding.

Here is how teachers and students can ask a question to AI at different stages of the learning process.

(1) Mathematics teachers can use AI in the following ways:

a) Lesson planning and content creation

- Generate practice activities: "Create 10 problems on quadratic equations with varying difficulty levels".
- Develop worksheets: "Make a worksheet on Pythagorean theorem with real-world applications".
- Create alternative explanations: "Explain complex numbers in 3 different ways for visual, auditory, and kinesthetic learners".
- Generate assessment items: "Create a quiz on derivatives with answer key and scoring rubric".

b) Differentiation and individualization

- Adapt problems: "Modify this calculus problem for a student who's struggling with algebraic manipulation".
- Create assignments: "Provide this geometry proof at basic, proficient, and advanced levels".
- Generate enrichment: "Create challenging problems about geometric sequences for advanced students".

c) Assessment and feedback

- Analyze common errors: "Identify common misconceptions in solving linear equations and suggest addressing strategies".
- Generate rubrics: "Create a grading rubric for the activities on quantifiers and their negation".
- Provide feedback templates: "Suggest specific, constructive feedback for a student who made a sign error in solving inequalities".

d) Professional development

- Stay current: "What are recent research-based strategies for teaching fractions?"
- Plan interventions: "Suggest interventions for students struggling with word problems".

- Designing projects to be assigned to students: "Help me design a project-based learning unit on exponential growth".

(2) Mathematics students can use AI in the following ways:

a) Personalized tutoring and explanation

- Get multiple explanations: "Explain the chain rule in calculus using simple expressions to be understood by a S4 student".
- Clarify concepts: "Why do we 'invert and multiply' when dividing fractions? Explain visually".
- Step-by-step guidance: "Walk me through solving this simultaneous equations step by step using matrices, but don't give me the answer".

b) Practice and problem-solving

- Generate similar problems: "Give me more problems like this one about probability".
- Check understanding: "Create a self-check quiz on trigonometric identities".
- Explore variations: "What if this geometry problem had different given information?"

c) Conceptual deepening

- Make connections: "How is the concept of slope related to real-world applications?"
- Explore why: "Why does the quadratic formula work? Derive it for me".
- Historical context: "What's the history behind the development of calculus?"

(3) Best practices and guidelines of using AI

For Teachers, use AI as a Co-planner, not replacement. Verify and edit AI-generated content for accuracy. Teach students HOW to use AI responsibly. Create AI-use policies for your classroom so that they focus on process over answers.

For Students, use AI as a tutor, not an answer key. Show your work and ask for feedback on process. Ask “why” and “how” not just “what’s the answer”. Do not copy AI solutions directly but use AI to understand methods. Do not depend solely on AI but use AI to supplement learning. Do not trust AI blindly but verify answers and methods.

In conclusion, the most effective mathematics classrooms will be those where teachers use AI to enhance their expertise and reach more students. Students use AI to get immediate, personalized support. Human connection between teachers and students remains central for motivation and deep understanding. This means that AI won’t replace mathematics teachers, but teachers who use AI effectively will replace those who don’t.

1.5.3 Human resource

The successful implementation of this syllabus relies heavily on qualified, competent, and motivated human resources. The following are key personnel required:

- **Teachers:** Professional and well-trained teachers with subject expertise and strong pedagogical skills in learner-cantered, participatory, and inclusive methods. Teachers should model integrity, tolerance, respect, responsibility, facilitate projects and apply effective assessment strategies.
- **Community resource persons:** Local bank managers, cooperative members, financial institutions, and professionals who share real-life experiences and act as mentors during classroom sessions or field visits.
- **Parents and guardians:** They are partners in education who provide moral support, monitor learner progress, and collaborate with teachers to reinforce entrepreneurial values, creativity, responsibility, and other competences at home, ensuring holistic learner development.
- **Local Authorities and Partners:** Support schools through linkages with business development services, innovation hubs, and community-based projects that offer practical learning opportunities.

Effective collaboration among stakeholders creates an enabling learning environment that equips learners with Mathematical knowledge, values, and practical competences needed for responsible citizenship, ethical behaviour, and meaningful contribution to national development.

Skills required for the Teacher of Mathematics

Effective implementation of this Syllabus requires teachers with strong subject knowledge, pedagogical skills, and interpersonal competence. As facilitators and role models, they guide learners to develop Mathematical mindsets and apply competences in real-life contexts. To achieve this, teachers should demonstrate the following key skills:

a) Subject matter expertise

- Strong mastery of entrepreneurship concepts such as innovation, business opportunity identification, marketing, finance, and sustainability.
- Ability to link entrepreneurship theory with real-life examples and national development priorities, including *Vision 2050* and *NST2*.
- Skill in relating Entrepreneurship to other disciplines such as Mathematics, ICT, Economics, and Social Studies.
- Awareness of current trends and technologies that shape the entrepreneurial landscape locally and globally.

b) Pedagogical and instructional skills

- Ability to design and implement learner-centred and participatory lessons that encourage active engagement, creativity, and critical thinking.
- Use of diverse teaching methods such as brainstorming, project-based learning, role-playing, case studies, and community engagement.
- Creativity and innovation in developing and applying diverse teaching resources, including ICT tools, realia, and community-based materials.
- Ability to adjust instruction to learners' levels, ensuring inclusivity and engagement for all.

c) Assessment skills

- Proficiency in designing and applying formative and summative assessments aligned with curriculum competencies.
- Effective use of rubrics, portfolios, checklists, and peer assessments to evaluate learners' competences and provide feedback that motivates learners and guides improvement.
- Skill in developing valid and fair test items that assess higher-order thinking and practical skills.
- Ability to analyze assessment results and use them to guide remedial or enrichment activities.

d) Classroom management and facilitation

- Ability to establish a respectful, safe, and inclusive learning environment that promotes tolerance and values diversity.
- Proficiency in managing group work, discussions, and collaborative learning activities.
- Consistent modelling of positive behaviour, respect, punctuality, and integrity as an example for learners.

e) Communication and interpersonal skills

- Strong oral and written communication to simplify complex ideas, stimulate dialogue, and foster critical reflection.
- Active listening and empathy to respond effectively to learners' academic and personal needs.
- Collaboration with colleagues, parents, school leaders, and community stakeholders to support holistic learner growth.
- Ability to motivate and inspire learners through encouragement and positive reinforcement.

f) Guidance and counselling skills

- Ability to provide moral, emotional, and motivational guidance to learners to build confidence, resilience, and ethical conduct.

- Demonstrating passion, patience, impartiality, and genuine care in dealing with learners' individual differences and challenges.
- Supporting learners in exploring business ideas, career paths, and responsible life choices.

g) Professional and ethical competence

- Upholding high ethical standards, integrity, impartiality in all teaching and assessment practices, and ethical conduct, serving as a role model for learners.
- Capacity to balance academic instruction with the cultivation of values and responsible citizenship.
- Demonstrating commitment to continuous professional development (CPD) through reflective practice, peer collaboration, and research.
- Modelling entrepreneurship values such as initiative, responsibility, and self-discipline.

h) Research and innovation skills

- Engagement in reflective practice and classroom-based research to improve teaching and learning.
- Adaptation of innovative approaches and new technologies to enhance learner participation.
- Connecting global entrepreneurship trends to Rwanda's socio-economic realities.
- Encouraging learners to use research findings in designing projects and solving community challenges.

Teachers who possess these competences will not only deliver Mathematics effectively but will also serve as role models who inspire learners to become creative, ethical, and productive contributors to Rwanda's development.

2. SYLLABUS UNITS

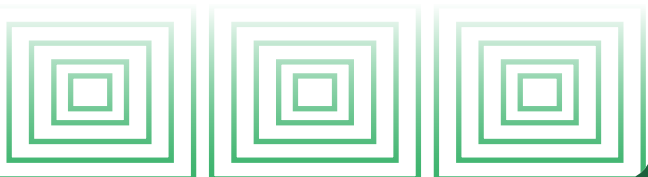
2.1 Presentation of the structure of the syllabus units

Basic Mathematics subject is taught and learnt in advanced level of secondary education as a **Minor Core Subjects** in S4, S5 and S6 respectively. At every grade, the syllabus is structured in topic areas, sub-topic areas where applicable and then further broken down into units to promote the uniformity, effectiveness and efficiency of teaching and learning Mathematics. The units have the following elements:

1. Each unit is aligned with the number of periods
2. Each unit has a Key Unit Competency the achievement of which is pursued by all teaching and learning activities undertaken by both the teacher and the learners
3. Each Key Unit Competence is broken into three types of Learning Objectives as follows:
 - a) *Type I*: Learning Objectives relating to Knowledge and Understanding (*Type I* Learning Objectives are also known as Lower Order Thinking Skills or LOTS)
 - b) *Type II* and *Type III*: These Learning Objectives relate to the acquisition of skills, attitudes and values (*Type II* and *Type III* Learning Objectives are also known as Higher Order Thinking Skills or HOTS) – These Learning Objectives are actually considered to be the ones targeted by the present reviewed curriculum.
4. Each Unit has content that indicates the scope of coverage of what is to be taught and learnt in line with the stated Learning Objectives
5. Each Unit suggests a non-exhaustive list of Learning Activities that are expected to engage learners in an interactive learning process as much as possible (learner-centred and participatory approach)
6. Finally, each Unit is linked to other subjects, the Assessment Criteria and the Materials (or Resources) that are expected to be used in the teaching and learning process.

The Basic Mathematics syllabus for Advanced level has got 4 Topic Areas: Algebra, calculus, Geometry, Statistics and Probability. As for units, they are 3 in S4, 4 in S5 and 3 in S6.

SENIOR FOUR



2.2 Basic Mathematics units for Senior Four

2.2.1 Key Competences at the end of Senior Four

After completion of Senior Four Basic Mathematics Topics, learners will be able to:

- Apply preliminarily mathematical concepts in solving real life problems.
- Use equations and inequalities to solve problems related to daily life.
- Analyze and interpret statistical data from daily life situations (different reference books, journals and relevant scientific documents, etc) and draw appropriate conclusions.

2.2.2 Syllabus units for Senior Four

Topic Area: Algebra			Sub-Topic: Numbers and Operations	
S4 Mathematics		Unit 1: Arithmetics		Number of periods: 33
Key unit competency: Use arithmetic operations in solving simple real-life problems				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe key concepts: sets, fractions, decimals, percentages, ratios, absolute value, powers, radicals, and logarithms. – Identify the properties of absolute value, powers, radicals, and logarithms.	– Perform operations (+, -, ×, ÷) with fractions, decimals, negative numbers, and radicals. – Convert between fractions, decimals, and percentages. – Solve real-life problems involving ratios, percentages, and absolute value.	– Appreciate the role of arithmetic in daily life and financial planning. – Value accuracy and precision in numerical calculations of final value of an investment.	– Introductory activity. – Negative numbers and related problems. – Decimals and related problems. – Fractions and related problems. – Ratios and related problems. – Percentages and related problems. – Absolute value (concept and properties).	– Pairs scenario-based: Learners work in pairs to solve real-life, scenario-based problems that involve calculating discounts using percentages, adjusting recipe quantities using ratios, and managing situations involving debt through the use of negative numbers.

– Differentiate simple from and compound interest. – Explain the concept of the arc of elasticity for demand. – Recall the formulas for final investment value and arc of elasticity for demand.	– Apply properties of powers and radicals to simplify expressions. – Rationalize the denominator of a radical expression. – Solve simple equations involving decimal logarithms. – Calculate simple interest, compound interest, and the final value of an investment. – Compute the arc of elasticity for demand and interpret the result.	– Demonstrate critical thinking and confidence when interpreting financial concepts like interest and elasticity.	– Problems involving absolute value. – Powers (concept and properties). – Problems involving powers. – Concept of radicals and properties. – Rationalization of radicals. – Problems involving radicals. – Concept of decimal logarithms and properties. – Simple equations involving decimal logarithms. – Simple interest and related problems. – Compound interest – Final value of investment. – Arc of elasticity for demand.	– Group activity: Learners engage in group activities to collaboratively simplify expressions involving powers, radicals, and rationalization accurately and efficiently. – Real life application: Students solve simple logarithmic equations to “decode” a secret message, reinforcing the concept of a logarithm as an exponent. – Group Project: Groups act as financial planners. They are given a client’s savings goal and must calculate the required principal, the difference between simple and compound interest, and the final investment value for different scenarios.
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				– Market Analyst: Groups are given data on price changes and demand for a product (e.g., coffee). They calculate the arc of elasticity, interpret whether demand is elastic or inelastic, and propose a pricing strategy.
Link to other subjects: Entrepreneurship (Calculate costs, revenue, profit, set prices for goods and services, prepare budgets and financial plans) and Economics (analyze numerical data such as GDP, inflation, and unemployment rates).				
Assessment criteria: Learners accurately and effectively apply the operations on real numbers in modeling and solving both mathematical and real-life problems.				
Teaching and learning materials: Graph papers, manila papers, digital technology including calculators and any AI tool.				

Topic Area: Algebra			Sub-Topic: Numbers and operations	
S4 Mathematics		Unit 2: Linear and quadratic equations and inequalities.		Number of periods: 33
Key Unit competence: Use linear and quadratic equations and inequalities in solving related problems from daily life situations.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Identify a linear equation, a quadratic equation, and an inequality. – Describe the concept of an interval and its properties.	– Solve linear equations and inequalities in one unknown. – Represent solutions of inequalities on a number line using interval notation.	– Value the accuracy and logical process required to solve equations. – Appreciate the power of algebraic methods in making informed decisions in economics and daily life.	Linear equations in one unknown: – Real life problems involving linear equations. Inequalities: – Meaning of inequalities. – Concepts of intervals and properties. – Problems involving intervals. – Inequalities involving products.	– Exploring Linear Equations through Context: Learners work in small groups to analyze real-life situation such as comparing the costs of two mobile phone plans, each with a base fee and a rate per minute. They form linear equations for each plan, solve them algebraically, and interpret the point of equality as the time when both plans cost the same. Groups then present their findings to the class, explaining their reasoning process.

– Explain the meaning of the discriminant and its role in determining the nature of the roots of a quadratic equation. – Identify the supply and demand equations in a market model. – State the different methods for solving simultaneous equations.	– Solve systems of two linear equations algebraically (substitution, elimination) and graphically. – Solve quadratic equations by factorization and by using the quadratic formula/discriminant. – Analyze and solve inequalities involving products and quotients.	– Demonstrate critical thinking when interpreting the feasibility of solutions in real-world contexts (e.g., a negative price is not feasible).	– Inequalities involving quotients. – Real life problems involving linear inequalities. Simultaneous linear equations and inequalities: – Solving algebraically two simultaneous linear equations by equating the same variables. – Solving algebraically two simultaneous linear equations by elimination method – Solving graphically two simultaneous linear equations. – Solving quadratic equations using discriminant. – Solving algebraically and graphically two simultaneous linear inequalities.	– Groups are given inequality problems (including products like $(x-2)(x+3)>0$). After solving, a representative places their group's solution interval on Manila paper. – Learners discuss overlaps and differences among solutions to reinforce understanding of intervals, products, and quotients in inequalities. – Modeling Real-Life Systems with Simultaneous Equations: Using examples from small business or market situations, groups are given linear equations representing supply and demand or cost and revenue. They solve the system algebraically using substitution and elimination, then verify graphically using graph paper or GeoGebra. Learners interpret the intersection point as the equilibrium price or break-even point and explain the meaning of profit and loss regions.
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	– Apply linear and quadratic models to solve real-life problems.		Quadratic equations: – Solving quadratic equations by factorization. – Concept of discriminant. – Applications of linear or quadratic equations: – Concept of supply and demand equations with examples. – Problems about supply and demand (equilibrium price), finance.	– Classifying Quadratic Equations through Discovery: Students are given a set of quadratic equations and, by calculating only discrimination, must quickly classify the nature of the roots (real, equal, imaginary) without solving fully. – Connecting Algebra to Everyday Decision-Making: Learners work collaboratively on short projects linking algebraic equations to real-life decisions, such as budgeting and financial planning. They present their models to the class, highlighting how algebra helps make rational and data-based decisions.
Link to other subjects: Economics: Supply and demand, break-even analysis, budgeting)				
Assessment Criteria: Learners accurately apply appropriate algebraic and graphical methods in solving linear and quadratic equations and inequalities, modeling and interpreting real-life situations accurately.				
Teaching and learning materials: Manila paper, graph paper, geometric instruments, ruler, model of geometric shapes in 2D, digital technology including GeoGebra, calculators, any AI tool, and projector.				

Topic Area: Statistics			Sub-Topic: Descriptive Statistics	
S4 Mathematics		Unit 3: Descriptive Statistics		Number of periods: 33
Key Unit competence: Analyze statistical data from daily life situations for drawing meaningful conclusions.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe measures of central tendency of data: mode, mean and median. – Distinguish between types of data. – Describe different methods of data presentation such as frequency tables, bar charts, pie charts, histograms, and polygons.	– Represent statistical information using: frequency distribution table, bar chart, pie chart, histogram, polygon. – Interpret graph of presented data. – Calculate and interpret measures of Centre tendency and measures of dispersion.	– Appreciate how data collection, data representation and data interpretation can be used for solving real life situations. – Appreciate the importance of measures of dispersion in the interpretation of data.	– Definition and type of data. – Presentation of data using frequency distribution table, bar chart, pie chart, histograms and polygons. – Graph interpretation. – Measures of central tendency for ungrouped data: Mode, Mean, Median. – Measures of central tendency for grouped data: mode, Mean, Median.	– In small groups, learners analyze a given dataset to calculate and interpret measures of dispersion (range, variance, and standard deviation) and present their results using visual aids such as charts or posters. – Learners use graph paper or digital tools (such as GeoGebra, Excel) to represent the data graphically, analyze the patterns or trends, and draw logical conclusions based on their observations.

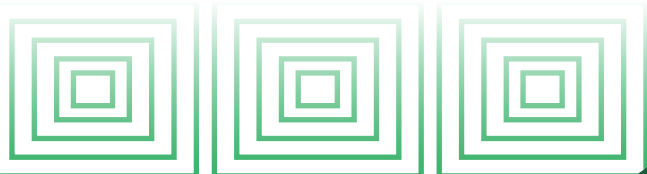
– Describe different methods of data presentation such as frequency tables, bar charts, pie charts, histograms, and polygons. – Explain the concepts and formulas for measures of central tendency and dispersion. – Interpret graphs and statistical results correctly.	– Display the distribution of data using the box plots. – Collect, represent, analyze and interpret data to answer a question.	– Show concern on how to use the standard deviation as measure of variability of data.	– Measures of dispersion: Quartiles, range, variance, Standard deviation, Coefficient of variation. – Application: Collection, organization, presentation and interpretation of data and draw conclusions.	– Learners carry out a mini-research task on a real-life problem (for example, students' study time, household expenses, or rainfall distribution). They collect, organize, and analyze data statistically, then communicate their findings through class presentations or written reports.
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Links to other subjects: *Geography (Assess population distribution variability across cities or regions, average population density), Finance (Determine average stock prices or returns over a period, Measure volatility of stock prices), Economics (Summarize GDP, GNP, or per capita income)*

Assessment criteria: *Learners accurately organize, represent and interpret real-life data using bar charts, pie charts and histograms, and determine measures of central tendency and dispersion.*

Teaching and learning materials: *Manila papers, Graph Papers, ruler, digital technology including GeoGebra, any AI tool, calculators*

SENIOR FIVE



2.3 Mathematics units for Senior Five

2.3.1 Key competences for end of Senior Five

After completion of Senior five Basic Mathematics, learners will be able to:

- Use logical statements to clearly express or convey information and make appropriate decisions.
- Represent geometric shapes in 2D and calculate their area.
- Apply graphical representation of functions to model problems related to daily life.
- Use arithmetic and geometric sequences to solve problems related to daily life situations.

2.3.2 Syllabus units for Senior Five

Topic Area: Algebra		Sub-topic area: Mathematical logic and applications		
S5: Mathematics	Unit 1: Introduction To Logic		Number of periods: 33	
Key Unit Competence: Use mathematical logic in reasoning and making decision in daily life.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe a simple statement, compound statement, and their truth values. – Explain the function of logical connectives: Negation, Conjunction, Disjunction, Conditional, and Biconditional.	– Differentiate between simple and compound statements. – Construct truth tables for all logical connectives and compound statements. – Convert everyday language statements into logical notation and vice versa.	– Show carefulness in distinguishing between factual statements and opinions. – Value precision in the use of logical language (e.g., "and" vs. "or").	– Introductory activity. – Simple statement. – Compound statements. – Truth values and Truth tables. – The negation "Not" – The conjunction "and" – The disjunction "or" – Concept of conditional statement "if ,..., then"	– Learners brainstorm sentences from daily life (news, ads, conversations) and classify them as simple/ non-statements and decide their possible truth values.

– Describe the concepts of converse, inverse, and contrapositive. – Identify tautologies and contradictions. – State the meaning of universal and existential quantifiers.	– Analyze a truth table to classify a statement as a tautology, contradiction, or contingency. – Apply rules to negate quantified statements correctly. – Demonstrate the validity of a hypothetical syllogism using logical principles. – Evaluate real-world arguments to identify valid reasoning or fallacies.	– Appreciate the role of clear definitions and logical structure in avoiding misunderstandings. – Develop a critical mindset towards unsubstantiated claims.	– Converse and contrapositive. – Inverse of a conditional. – The bi-conditional statement "if and only if" – Tautology. – Contradiction. – Quantifiers: Universal quantifier "for all" – Existence quantifier "There exists" – Negation of quantifiers. – Hypothetical syllogism. – Affirming the antecedent. – Denying the consequent.	– Using cards with simple statements, learners physically build compound statements using connectives ("and", "or", "if... then") and predict the truth value under different scenarios. – Learners are given worksheets to systematically construct truth tables for increasingly complex statements, culminating in identifying a tautology and a contradiction.
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				– Learners practice translating English sentences with "all" and "some" into logical quantifiers and then correctly negating them (e.g., "All birds fly" → "Not all birds fly" or "Some birds do not fly").
Link to other subjects: Physics (Analyzing cause-and-effect relationships : e.g., if force increases, then acceleration increases), Psychology (Analyzing arguments and identifying fallacies), Literature (Making argumentative and persuasive writing) Economics (Economic theories and decision models rely on logical reasoning :e.g., If price increases, then demand decreases).				
Assessment criteria: Learners apply mathematical logic in reasoning effectively and making sound decisions in everyday life situations.				
Teaching and learning materials: Manila paper, markers, colored cards				

Topic Area: Geometry			Sub-topic area: Plane Geometry	
S5: Mathematics		Unit 2: Point, Lines and Geometric Shapes In 2D		Number of periods: 21
Key Unit Competence: Apply 2D shapes properties in calculating the area of land.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Describe coordinate of a point in 2D. – State properties of a straight line.	– Represent a point and a vector in 2D. – Calculate the distance between two points in 2D and the mid - point of a segment in 2D. – Calculate the angle between two straight lines.	– Appreciate that a point is a fixed position in a plane. – Be accurate in representations and calculations.	– Introductory activity – Cartesian coordinates of a point in 2D. – Distance between two points in 2D. – Mid-point of a line segment. – Distance between two points in 2D. – Vector in 2D. – Dot product of vectors in 2D.	– Use masking tape to create a large Cartesian plane on the classroom floor. Students become “points.” Call out coordinates (e.g., A (3, 4), B (-2, 1)) and students must stand on the correct location. Students find the distance between themselves and a partner using the distance formula. – In the “Human Coordinate Plane,” two students represent two friends’ houses at specific coordinates. A third student must find the “mid-point” where they should meet to walk equal distances.

– Provide conditions of parallelism and perpendicularity of two straight lines. – Classify the geometric shapes in 2D.	– Determine the equations of a straight line (parametric and cartesian equation). – Determine the intersection of lines. – Calculate the perimeter and area of geometric shapes in 2D.	– Manifest a satisfaction that an angle is made by the intersection of two straight lines. – Appreciate the importance of determining perimeter and area in real life.	– Properties of dot product in 2D – Vector equation of a straight line. – Parametric equation of a straight line – Cartesian equation of a straight line – Position of a point to the line. – Position of two lines – Condition of parallelism. – Angles between two lines. – Condition of perpendicularity. – Identification of Geometric shapes in 2D.	– Using a ruler, students draw a line on graph paper. They identify two points on the line, calculate the slope, and derive the Cartesian equation ($y = mx + c$) or ($ax + by + c = 0$). task is to subdivide it into simpler shapes (parallelograms, pentagons, hexagons, triangles) to calculate the total area. They then compare lines with different slopes and intercepts to see how the equation affects the line's steepness and position. – Students are given one line (a railroad track) and a point where a new, parallel track must pass. They must find the equation of the parallel line. They discover that parallel lines have equal slopes ($m_1 = m_2$).
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			– Perimeter and area of geometric shapes in 2D: Square, Triangle, Rectangle, parallelogram and trapezium. – Perimeter and area of geometric shapes in 2D: Parallelogram, Pentagon and Hexagon.	– Students are given a checklist of shapes (square, rectangle, various triangles, parallelogram, pentagon, hexagon). They must find and photograph or sketch real-world examples of these shapes around the school (e.g., a window is a rectangle, a yield sign is a triangle). – Students act as land surveyors. They are given a large, irregular piece of land on grid paper (e.g., a composite shape). Their task is to subdivide it into simpler shapes (parallelograms, pentagons, hexagons, triangles) to calculate the total area.
Link to other subjects: Physics: <i>Understanding force diagrams, vectors, and motion paths.</i> Geography: <i>Map reading uses coordinate systems (latitude and longitude) similar to Cartesian coordinates.</i>				
Assessment criteria: <i>Learners apply appropriate mathematical formulas and equations in representing points, vectors, and lines in 2D, determining distances, midpoints, angles, and relationships between lines, and calculating the perimeter and area of geometric shapes accurately.</i>				
Teaching and learning materials: <i>Manila paper, graph paper, geometric instruments, model of geometric shapes in 2D, digital technology including GeoGebra, any AI tool, calculators, computer, projector and internet.</i>				

Topic Area: Analysis			Sub-topic area: Functions	
S5: Mathematics	Unit 3: Graphs and Functions		Number of periods:21	
Key Unit Competence: Apply graphical representation of functions in solving economics and financial problems.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Determine the domain and range of a function. – Perform operations on functions. – Describe constant and identity functions and recognize their properties.	– Draw graphs that corresponds to linear, non-linear functions; – Find the slopes of linear functions and tangents to non-linear function by graphical analysis. – Use the slope of a linear demand function to calculate point elasticity.	– Showing interest in exploring how one quantity depends on another in real-life situations. – Valuing precision when determining whether a rule truly defines a function.	– Generalities on numerical functions. – Constant function and identity functions. – Monomial and polynomial functions. – Rational and irrational functions. – Domain of definition of polynomial functions.	– Introduce the definition of a function using real-life examples (e.g., price vs. quantity, distance vs. time). – Demonstrate how to determine the domain of polynomial, rational, and irrational functions and learners work in pairs to find and explain domains of sample functions.

– Explain monomial, polynomial, rational, and irrational functions and state their general forms. – Explain how to determine the domain of definition for polynomial, rational, and irrational functions. – Distinguish between even and odd functions and explain their graphical symmetry properties. – Explain the concept of the inverse of a function and conditions for its existence.	– Analyze, model and solve problems involving linear or quadratic functions and interpret the results. – Interpret the meaning of functions with two independent variables: marginal revenue and marginal cost functions. – Determine the domain and range of various types of functions (polynomial, rational, irrational). – Perform operations on functions (addition, subtraction, multiplication, division, composition) correctly.	– Appreciating logical methods used to distinguish valid functions from non-functions. – Demonstrating patience in analyzing and testing values to find valid inputs and outputs. – Increase self-confidence and determination to appreciate and explain the importance of functions – Evaluate cost and profit situations using mathematical reasoning to support decision-making.	– Domain of definition of rational functions. – Domain of definition of irrational functions. – Parity of a function (odd or even). – Addition and subtraction of functions. – Multiplication and division of functions. – Composite function. – Inverse of a function. – Graphical representation and interpretation of linear function. – Graphical representation and interpretation of quadratic function.	– Introduce operations on functions (addition, subtraction, multiplication, division, composition) step-by-step. – Use mapping diagrams and tables to visualize relationships. – Explore and graph each type of function using graphing tools or graph paper. – Guide learners through finding values that make the function undefined. Use simple examples (e.g., division by zero, negative roots) for hands-on analysis.
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– Describe the graphical representation and key characteristics of linear and quadratic functions. – Explain the economic meaning of functions such as cost, revenue, profit, price, and consumption functions.	– Identify and verify the inverse of a function through algebraic and graphical methods. – Identify whether a function is even or odd using algebraic and graphical methods. – Calculate and interpret marginal cost, marginal revenue, and marginal profit from given data or functions.	– Apply teamwork, discussion, and communication skills when exploring graphical and applied problems. – Appreciate the usefulness of mathematical models in describing and predicting real-world phenomena.	– Functions in economics: • Price as function of quantity supplied and consumption as function of income, • Price as function of quantity demanded and The Cost Function, • The Revenue Function and The Profit Function, • The Marginal Cost, Marginal Revenue, and Marginal Profit.	– Use simple examples to show how one function's output becomes another's input. – Plot graphs of $y = mx + c$ and $y = ax^2 + bx + c$ manually and digitally. – Work collaboratively to solve applied function problems and present findings to the class.
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Link to other subjects: **Physics** (Describing distance, time, velocity, time,), **Economics** (Describing of demand and supply, ...). **Geography** (Representing and analyzing geographical data such as population growth, rainfall, ...) **Biology** (modeling relationships in biological systems such as enzyme activity vs. temperature, or population vs. time), **Chemistry** (Representing gas laws, reaction rate vs. Temperature, ...) **Accounting** (in representing sales growth or depreciation).

Assessment criteria: Learners effectively apply functions in modeling, analyzing, and solving real-life problems.

Teaching and learning materials: Examples from daily life like price-quantity tables, production data, and population growth chart, s Graph papers, ruler, digital resources (including calculators, GeoGebra software and any AI tool).

Topic Area: Algebra			Sub-topic area: Number Patterns	
S5: Mathematics		Unit 4: Sequences		Number of periods: 24
Key Unit Competence: Apply the concepts and formulas of arithmetic and geometric sequences in solving real-life problems involving patterns and progression.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Define a sequence, arithmetic sequence, and geometric sequence. – State the formulas for the general term and the sum of the first n terms. – Explain the concepts of arithmetic and geometric mean.	– Identify whether a given sequence is arithmetic, geometric, or neither. – Calculate the general term, specific terms, and the sum of the first n terms of a sequence. – Solve for unknown variables in sequence problems (e.g., first term, common difference/ratio, number of terms).	– Appreciate the usefulness of sequences in modeling real-life phenomena. – Develop patience and logical reasoning when working through multi-step sequence problems. – Value the role of sequences in planning and predicting future outcomes (e.g., savings, construction).	– Definition of sequences. – Arithmetic sequences: • General term of an arithmetic sequence. • Arithmetic mean. Sum of n first terms of an arithmetic sequence.	– Teacher facilitates a brainstorming session. Provides groups activities with different number patterns. Guides the class towards a formal definition and the learners in groups, analyze number sets, describe patterns, predict next terms, and create a simple rule. Share findings with the class. – Teachers presents scenarios showing multiplicative change. Introduces the general term and sum formulas.

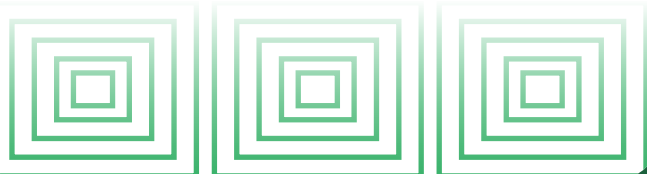
– Describe the general term of arithmetic and geometric sequences. – Identify real-life situations that can be modeled by arithmetic or geometric sequences.	– Solve for unknown variables in sequence problems (e.g., first term, common difference/ratio, number of terms). – Apply sequences to model and solve real-world problems (financial, population growth, etc.). – Solve real-life problems using arithmetic and geometric sequences.	– Show accuracy, logical reasoning, and persistence when calculating terms and sums of sequences. – Develop confidence and perseverance in solving both theoretical and applied sequence problems.	– Geometric sequences: • General term of a geometric sequence. • Geometric mean. Sum of n first terms of a geometric sequence. – Application of sequences in solving real life problems.	Guides the algebraic derivation of the sum formula then the learners analyze growth/decay scenarios. Practice using the general term. Derive the sum formula through guided steps and practice its application. – Teachers provides a “case file” of real-world problems. Circulates, provides hints, and checks for correct reasoning and learners work in groups as “problem solvers.” Analyze each scenario, justify the sequence type, and apply the correct formulas to find solutions.
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Link to other subjects: *Physics* (to analyze motion and change in physical quantities), *Economics* (Using a sequence model financial growth and depreciation over time), *Geography* (Analyzing population growth, migration trends, or resource depletion over time), *Biology* (Understanding exponential growth and decay), *Chemistry* (Reactions involving half-life decay or concentration decrease can be expressed using geometric sequences).

Assessment criteria: Learners correctly apply formulas in determining the general term, specific terms, arithmetic/geometric means, and the sum of the first n terms of a sequence.

Teaching and learning materials: Graph papers, manila papers, digital technology including calculators

SENIOR SIX



2.4 Mathematics syllabus units for Senior Six

2.4.1 Key competences for end of Senior Six

After completion of Senior Six Basic Mathematics Topics, learners will be able to:

- Extend understanding, analysis and interpretation of bivariate data to correlation coefficients and regression lines
- Use counting techniques to determine the number of possible outcomes of events occurring in real life.
- Determine the probability of possible outcomes of events occurring under equally likely assumptions.

2.4.2 Syllabus units for Senior Six

Topic Area: Statistics And Probability			Sub-topic area: Descriptive Statistics	
S6: Mathematics		Unit 1: Two- Variable Statistics		Number of periods: 36
Key Unit Competence: Interpret two-variable data using correlation and regression techniques in making predictions and drawing conclusions.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain correlation and list its different types (positive, negative, zero). – Explain the concept of covariance and its role in measuring the direction of a linear relationship. – Recall the formula for the standard deviation of a variable.	– Construct and accurately plot a scatter diagram from a set of bivariate data. – Calculate the covariance for a given set of bivariate data. – Calculate the standard deviation for individual variables in a dataset.	– Appreciate the importance of statistical methods in describing relationships in the real world. – Develop a critical attitude towards interpreting statistical results, understanding that correlation does not imply causation.	– Introductory activity. – Concept of bivariate data. – Scatter diagram plotting. – Concept of Correlation and examples. – Types of correlation. – Description of Covariance.	– Engage in an introductory activity involving real-life paired data to discuss possible relationships. – Collect and organize two-variable data from their environment – Demonstrate how to plot scatter diagrams and interpret the type and strength of correlation visually. – Explain and illustrate the concept of correlation, covariance, and their types with simple numerical examples in group.

– State the concept, formula, and key properties of the Pearson's Product-Moment Correlation Coefficient. – Recall the equation of a straight line ($y = mx + c$) and its relation to the concept of a regression line. – Recall the relationship between the equation of a line, gradient, and regression. – Explain the concept of prediction (interpolation and extrapolation) using a regression line.	– Calculate Pearson's Correlation Coefficient. – Determine the equation of the regression line of y on x from given summary statistics or raw data. – Use the regression line to make predictions (estimate y for a given x). – Analyze and interpret statistical output (scatter plots, correlation coefficients, contexts)	– Value accuracy and meticulousness in calculations and data plotting. – Develop a collaborative spirit when working on data analysis projects with peers. – Demonstrate a willingness to use statistical evidence to support arguments and make informed decisions.	– Determination of covariance for bivariate statistics data. – Review on determination of standard deviation. – Concept and properties of coefficient of Person's correlation. – Problems involving Pearson's coefficient of correlation. – Review on equation of a line defined by a point and a gradient. – Concept of regression line.	– Class discussions to compare positive, negative, and zero correlations using real-life scenarios. – Provide exercises for computing covariance and standard deviation from given bivariate data sets. – Determine the coefficient of correlation (Pearson's) and interpret results. – Review the equation of a line defined by a point and gradient and connect it to the concept of a regression line. – Demonstrate how to determine regression lines (y on x) and use them for prediction in practical contexts (business, science, or social studies).
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	regression equations) to draw meaningful conclusions in various. – Evaluate the relationship between variables and the strength of a correlation.	– Recognize the ethical responsibility in presenting and interpreting statistical findings, avoiding misleading conclusions.	– Regression line of y on x and prediction – Problems involving regression line and prediction in various areas. – Problems involving data analysis and interpretation in various areas.	– Solve problems with tasks requiring data analysis, interpretation, and prediction in various applied fields. – Facilitate peer discussion and presentation of results to promote collaboration and critical thinking. – Conclude with a reflection session where learners discuss the importance of correlation and regression in decision-making and research.
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Link to other subjects; *Economics (income and expenditure, price and demand, or investment and profit). Geography (rainfall and crop, altitude and temperature, or population and resources). Biology (age and heart rate, temperature and enzyme activity, or dosage and response). Chemistry (concentration and reaction rate, temperature and solubility, or energy and wavelength). Agriculture (fertilizer quantity and crop yield, or rainfall and production levels).*

Assessment criteria: *Learners accurately predict relationships between two related variables using concepts of correlation and regression in real-life situations.*

Teaching and learning materials: *Graph papers and rulers for plotting scatter diagrams, Calculators or computers with spreadsheet software (e.g., Excel), any AI tool, Sample data sets from real-life contexts (e.g., weather, business, sports, population)*

Topic Area: Statistics and Probability			Sub-topic area: Probability	
S6: Mathematics		Unit 2: Counting Techniques		Number of periods: 36
Key Unit Competence: Apply counting techniques in determining the number of possible outcomes from the given events.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain key terms: permutations, combinations, mutually exclusive events, and indistinguishable objects. – Describe the purpose and properties of Venn diagrams, tree diagrams, and contingency tables.	– Construct tree diagrams and contingency tables to model real-world scenarios. – Apply the Product and Sum Principles to calculate outcomes for multi-stage and mutually exclusive events.	– Appreciate the usefulness of counting techniques in solving real-life problems. – Develop logical reasoning and accuracy when applying counting principles.	– Use of Venn diagram as counting techniques and related properties. – Concept of tree diagrams. – Determination of number of outcomes using tree diagrams. – Concept of contingency table and properties – Determination of number of outcomes using contingency table. – Description of product principle. – Determination of number of outcomes using Product principle.	– Review basic set concepts and demonstrate the use of Venn diagrams in counting. Show examples of union, intersection, and complement. – Construct tree diagrams for given experiments (e.g., tossing two coins, picking fruits). Count and record all possible outcomes. – Calculate number of possible outcomes using their drawn tree diagrams. Compare with classmates' results.

– State the Fundamental Product Principle and Sum Principle. – Explain the conceptual difference between a permutation and a combination. – Recall the standard formulas for permutations, and circular arrangements. – Classify a given counting problem as a direct application of the sum principle, product principle, permutation.	– Calculate the number of permutations for linear and circular arrangements, including cases with indistinguishable objects. – Analyze complex counting problems by breaking them down into simpler parts that utilize the appropriate principles.	– Show persistence and curiosity in exploring different ways of arranging and selecting objects. – Value organization, precision, and systematic thinking in mathematical problem-solving. – Evaluate the most efficient counting technique for a given problem.	– Concept of Mutually exclusive events and examples. – Concept of sum principle as counting technique and related problems. – Problems involving sum and product principles. – Permutations of indistinguishable objects: Permutations with repetitions of n objects chosen from n objects. – Distinguishable arrangements (Permutations of r unlike objects chosen from n distinct objects without repetition). – Indistinguishable arrangements: Permutations of r objects selected from mixture of n alike and unlike objects.	– Introduce contingency tables, showing how data can be arranged in rows and columns. Use classroom data as example. – Complete given contingency tables and calculate overall totals. Discuss results in small groups. – Explain product principle using real-life contexts (e.g., clothing choices). Relate to multiplication rule in probability. – Solve exercises involving multiple choices and verify by drawing tree diagrams. – Define and illustrate mutually exclusive events using Venn diagrams. Contrast with overlapping events. – Teacher uses some videos in illustrating mutually exclusive events.
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– Illustrate the difference between a linear permutation and a circular permutation with a concrete example. – Summarize the logical reasoning behind the formula for permutations of indistinguishable objects.	– Solve advanced problems involving mixtures of alike and unlike objects.	– Defend their chosen solution path by justifying the use of a specific principle or formula. – Design a systematic approach to solve novel, multi-step counting problems. operational planning.	– Ordered samples and the product rule of counting. – Circular permutation of all n objects. – Combinations: Definitions and properties. – Combinations with repetitions or conditions. – Application of counting techniques.	– Solve problems requiring the use of sum principle and explain why addition applies. – Work in pairs to solve mixed problems involving both principles and justify solutions. – Illustrate properties of permutations and practice simplification using factorials. – Work in groups to solve real-life problems involving counting techniques and present their findings to class.
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Link to other subjects; Economics (in market choices, product arrangements, and probability of economic outcomes). **Geography** (Calculating combinations of geographical features or population groupings). **Biology** (In genetics to determine possible combinations of genes and traits).

Assessment criteria: Learners accurately apply the principles of counting techniques in interpreting the number of possible outcomes in various contextual and problem-solving situations.

Teaching and learning materials: Dice, coins, cards, manila paper, graph paper, ruler, digital technology including calculators, videos for games, and any AI tool.

Topic Area: Statistics And Probability			Sub-topic area: Probability	
S6: Mathematics		Unit 3: Elementary Probability		Number of periods: 27
Key Unit competence: Determine the probability of possible outcomes of events occurring under equally likely assumptions to inform in decisions making.				
Learning Objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and values		
– Explain key concepts of probability (experiment, sample space, event). – Describe and distinguish between mutually exclusive, inclusive, independent, and dependent events.	– Determine the sample space and events for a given experiment. – Compute the probability of simple and compound events using appropriate rules. – Use tree diagrams to represent and calculate probabilities.	– Appreciate the role of probability in decision-making and real-life problem solving. – Demonstrate accuracy, logical reasoning, and critical thinking when solving probability problems.	– Concept of probability (the probability, experiment, sample space and event) and examples. – Probability of an event. – Basic probability rules. – Probability of mutually exclusive (incompatible). – Probability of inclusive events and additive rule.	– In groups, discuss the advantages and disadvantages of betting and present the findings to the class. – Explain key terms: experiment, sample space, and event using visual aids and practical demonstrations. – Demonstrate how to calculate the probability of simple and compound events using examples on the board.

– State and apply basic probability rules (addition and multiplication rules). – Explain the concept of conditional probability. – Identify the relationship between arrangements, combinations, and probability. – Recognize real-life situations that can be modeled using probability.	– Apply the addition and multiplication rules to solve problems involving independent and dependent events. – Calculate conditional probabilities in real-life contexts. – Solve probability problems involving arrangements and combinations. – Interpret and analyze real-life situations using probability concepts.	– Develop persistence and confidence in tackling uncertainty in mathematical tasks. – Show curiosity and interest in exploring real-world applications of probability. – Value fairness and objectivity when interpreting random events and data outcomes.	– Concept of independent events and examples. – Probability of independent events. – Multiplication rule. – Multiplication rule using tree diagram. – Concept of dependent events and examples. – Probability of dependent events (Conditional probabilities). – Probabilities involving arrangements and combinations. – Examples of real-life events and determination of related probability.	– Guide learners to distinguish between mutually exclusive, inclusive, independent, and dependent events through examples and class discussion. – Construct and interpret tree diagrams to find probabilities of combined or conditional events. – Present real-life tasks (e.g., weather prediction, games, traffic flow, or card games) and ask learners to determine related probabilities. – Facilitate a class discussion on the importance of probability in decision-making and risk management.
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Links to other subjects: Biology (Used to model genetic inheritance (Mendel's laws), population dynamics, and mutation rates), Chemistry (Probability models reaction rates, molecular collisions, and random decay), Physics (Used in statistical mechanics, quantum theory, and thermodynamics to model randomness and uncertainty), Language (Linguists use probabilities to study phoneme frequency, word distributions, and syntax patterns).

Assessment criteria: Apply concepts of counting techniques in solving real-life problems involving arrangement, selection, and organization of objects or events.

Resources: Dice, coins, cards, Manila paper, graph paper, ruler, digital technology including calculators, videos for games, and any AI tool.

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ANNEX: TIME ALLOCATION PER SUBJECT IN CONCERNED PATHWAYS

A) LANGUAGES PATHWAY

Subject	Number of periods		
I. Major Core subjects/ Examinable at National Level	(Period= 40 minutes)		
	S4	S5	S6
1. English	7	7	7
2. French	7	7	7
3. Kinyarwanda	7	7	7
4. Kiswahili	7	7	7
II. Minor Core Subjects/ Examinable at National Level			
5. History	3	3	3
6. Psychology	3	3	3
7. Basic Mathematics	3	3	3
Sub-total 1	37	37	37
III. General Subjects: Examinable at School level			
8. ICT	3	3	3
9. Entrepreneurship	3	3	3
10. General and religious Studies	3	3	3
11. Physical Education and Sports	2	2	2
Sub-total 2	11	11	11

III. Co-curricular activities (compulsory)	2	2	2
Total number of contact periods per week	50	50	50
Total number of contact hours/week	33.3	33.3	33.3
Total number of contact hours per year (39 weeks)	1300	1300	1300

B) ARTS AND HUMANITIES PATHWAY

Subject	Number of periods		
I. Major Core Subjects: Examinable at National Level	(Period= 40 minutes)		
	S4	S5	S6
1. History	7	7	7
2. Literature in English	7	7	7
3. Geography	7	7	7
4. Psychology	7	7	7
II. Minor Core Subjects: Examinable at National Level			
5. French	3	3	3
6. Kinyarwanda	3	3	3
7. Basic Mathematics	3	3	3
Sub-total 1	37	37	37

III. General Subjects: Examinable at School level			
8. ICT	3	3	3
9. Entrepreneurship	3	3	3
10. General and Religious Studies	3	3	3
11. Physical Education and Sports	2	2	2
Sub-total 2	11	11	11
III. Co-curricular activities (compulsory)	2	2	2
Total number of contact periods per week	50	50	50
Total number of contact hours/week	33.3	33.3	33.3
Total number of contact hours per year (39 weeks)	1300	1300	1300